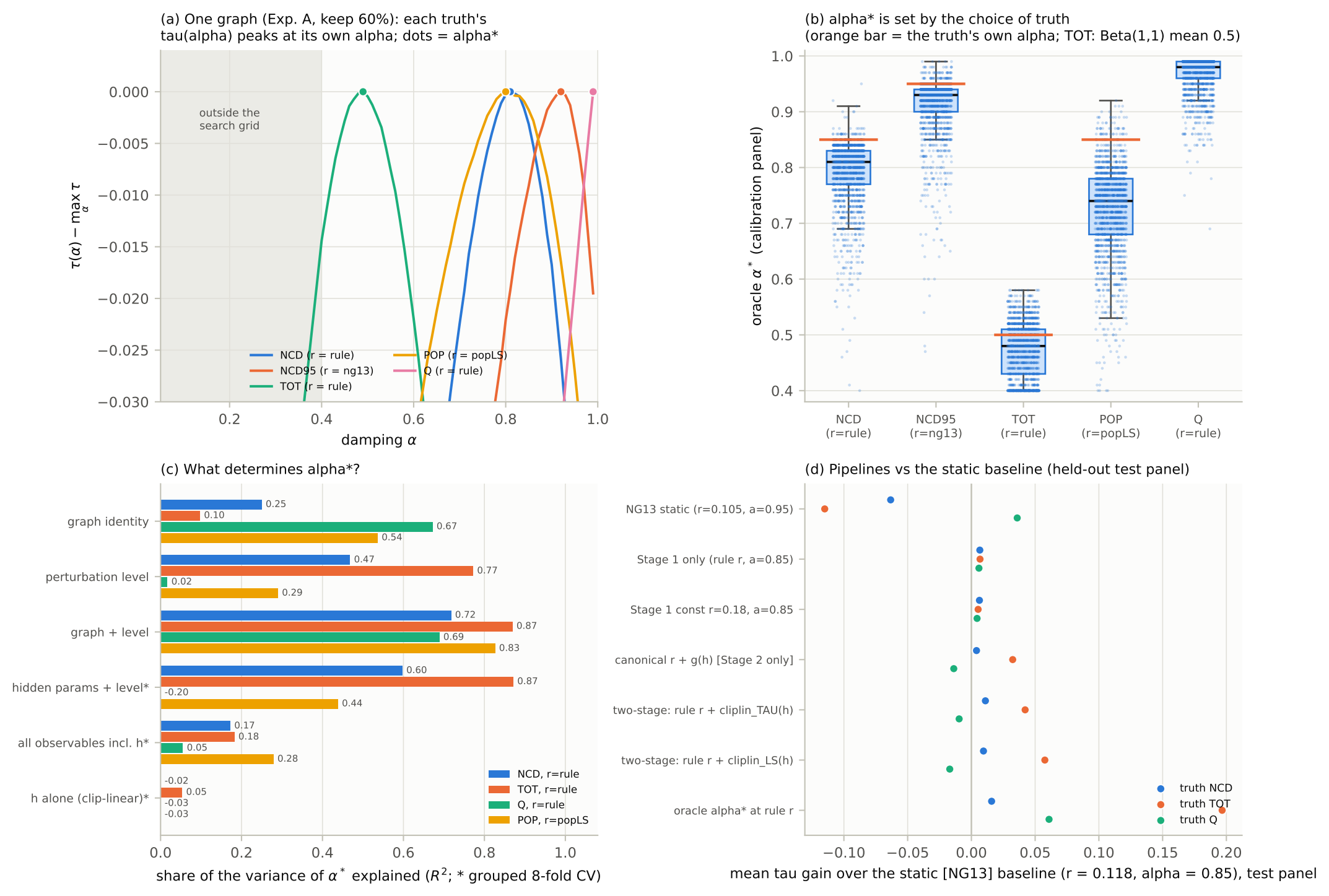

(a) One graph (Exp. A, keep 60%): each truth's tau(alpha) peaks at its own alpha; dots = alpha*
outside the search grid
NCD (r = rule)
NCD95 (r = ng13)
TOT (r = rule)
POP (r = popLS)
Q (r = rule)
damping α
(b) alpha* is set by the choice of truth
(orange bar = the truth's own alpha; TOT: Beta(1,1) mean 0.5)
oracle α* (calibration panel)
NCD (r=rule)
NCD95 (r=ng13)
TOT (r=rule)
POP (r=popLS)
Q (r=rule)
(c) What determines alpha*?
graph identity
perturbation level
graph + level
hidden params + level*
all observables incl. h*
h alone (clip-linear)*
NCD, r=rule
TOT, r=rule
Q, r=rule
POP, r=popLS
share of the variance of α* explained (R²; * grouped 8-fold CV)
(d) Pipelines vs the static baseline (held-out test panel)
NG13 static (r=0.105, a=0.95)
Stage 1 only (rule r, a=0.85)
Stage 1 const r=0.18, a=0.85
canonical r + g(h) [Stage 2 only]
two-stage: rule r + cliplin_TAU(h)
two-stage: rule r + cliplin_LS(h)
oracle alpha* at rule r
truth NCD
truth TOT
truth Q
mean tau gain over the static [NG13] baseline (r = 0.118, alpha = 0.85), test panel